

Assignment - 10

Part - A

1. State Taylor's theorem

If a fn $f(z)$ is analytic inside and on a circle centered at a , the $f(z)$ can be expanded as a power series about $z=a$

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!}f''(a) + \dots$$

or

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n$$

2. State Laurent's theorem

If $f(z)$ is analytic in an annular region

$$R_1 < |z-a| < R_2$$

then $f(z)$ can be expanded as

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=0}^{\infty} \frac{b_n}{(z-a)^n}$$

3. State Cauchy's Residue theorem.

If $f(z)$ is analytic inside and on a single closed contour C , except at isolated singularities inside C , then

$$\oint_C f(z) dz = 2\pi i \times (\text{sum of residues inside } C)$$

i.e

$$\oint_C f(z) dz = 2\pi i \sum \text{Res}(f, a_n)$$

4 Define Isolated singularity and obtain the poles of $\frac{z^3}{(z-2)(z-3)}$

Ans A point $z=a$ is called an isolated singularity of $f(z)$ if $f(z)$ is not analytic at a but is analytic in some neighbourhood around a except at the point itself

Given $f(z) = \frac{z^3}{(z-2)(z-3)}$

Singularities occur where denominator becomes zero.

$$(z-2)(z-3) = 0$$

$$\boxed{z=2, 3}$$

Both are simple poles because each factor occurs once.

5 Determine the poles of the function $\frac{z^2}{(z-a)(z-b)(z-c)}$

$$\frac{z^2}{(z-a)(z-b)(z-c)}$$

Denominator becomes 0.

$$z=a, z=b, z=c \rightarrow \text{poles}$$

Part - B

1. Expand $f(z) = \sin z$ in Taylor's Series about $z = \pi/4$

Taylor expansion about $z = a$

$$f(z) = f(a) + (z-a)f'(a) + \frac{(z-a)^2}{2!}f''(a) + \dots$$

$$a = \pi/4$$

$$f(z) = \sin z$$

$$f'(z) = \cos z$$

$$f''(z) = -\sin z$$

$$f'''(z) = -\cos z$$

$$@ \pi/4 \quad \sin \pi/4 = \cos \pi/4 = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\sin z = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}(z - \frac{\pi}{4}) - \frac{1}{\sqrt{2}} \left(\frac{(z - \pi/4)^2}{2!} \right) - \frac{1}{\sqrt{2}} \left(\frac{(z - \pi/4)^3}{3!} \right) + \dots$$

$$\sin z = \frac{1}{\sqrt{2}} \left[1 + (z - \pi/4) - \frac{(z - \pi/4)^2}{2!} - \dots \right]$$

2. Determine poles and residue of $f(z) = \frac{z^2}{(z-1)^2(z+2)}$

and evaluate

$$\oint_C \frac{z^2 dz}{(z-1)^2(z+2)}, \quad |z| = 3.$$

find poles

$$(z-1)^2(z+2) = 0$$

$z = 1$ (pole of order 2)

$z = -2$ (simple pole)

Residue @ $z = -2$

for single pole:

$$\text{Res}(f, -2) = \lim_{z \rightarrow -2} (z+2) \frac{z^2}{(z-1)^2(z+2)}$$

$$= \lim_{z \rightarrow -2} \frac{z^2}{(z-1)^2}$$

$$= 4/9$$

Residue @ $z = 1$

for pole of order 2:

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} \frac{d}{dz} \left(\frac{z^2}{z+2} \right)$$

$$\frac{d}{dz} \left(\frac{z^2}{z+2} \right) = \frac{2z(z+2) - z^2}{(z+2)^2}$$

$$@ z = 1 \quad \frac{1+4}{9} = \underline{\underline{5/9}}$$

Apply residue Theorem

$$\frac{4}{9} + \frac{5}{9} = 1$$

$$\oint f(z) dz = 2\pi i (\pm)$$

Residues are $\frac{5}{9}$ and $\frac{4}{9}$

$$\oint \frac{z^2 dz}{(z-1)(z+2)} = 2\pi i$$

3 Expand

$$f(z) = \frac{1}{z^2 - 3z + 2} \quad \text{valid for the regions}$$

$$(i) |z| < 1$$

$$(ii) |z| > 2$$

factor denominator

$$z^2 - 3z + 2 = (z-1)(z-2)$$

Partial fractions

$$\frac{1}{(z-1)(z-2)} = \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

$$1 = A(z-2) + B(z-1)$$

$$1 = Az - 2A + Bz - B$$

$$A + B = 0 \quad \times \quad 2$$

$$-2A - B = 1$$

$$2A + 2B = 0$$

$$-2A - B = 1$$

$$\boxed{B = 1, A = -1}$$

$$\frac{-1}{z-1} + \frac{1}{z-2} = \frac{1}{(z-1)(z-2)}$$

in region $|z| < 1$ $\frac{1}{|z-1|} = \frac{1}{1-z}$

$$= \sum_{n=0}^{\infty} z^n$$

$$\frac{1}{z-2} = -\frac{1}{2} \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$$

$$\therefore f(z) = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n$$

(ii) region $|z| > 2$
write power of $1/z$

$$\frac{1}{z-1} = \frac{1}{z} \frac{1}{1-1/z} = \sum_{n=0}^{\infty} \frac{1}{z^{n+1}}$$

Thus, $f(z) = \sum_{n=0}^{\infty} \frac{z^n - 1}{z^{n+1}}$

part - c

Q. 9 Find the Laurent expansion of $f(z) = \frac{1}{(z^2+1)(z+2)}$ when $|z| > 2$ and $1 < |z| < 2$.

$$\frac{1}{(z^2+1)(z+2)} = \frac{Az+B}{z^2+1} + \frac{C}{z+2}$$

$$1 = (Az+B)(z+2) + C(z^2+1)$$

Solving we get, $A = -\frac{1}{5}$, $B = \frac{2}{5}$, $C = \frac{1}{5}$

Hence
$$f(z) = \frac{-z+2}{5(z^2+1)} + \frac{1}{5(z+2)}$$

① Laurent expansion for $|z| > 2$

~~expand~~ $\frac{1}{z+2}$

$$\frac{1}{z+2} = \frac{1}{z} \cdot \frac{1}{1+\frac{2}{z}} = \frac{1}{z} \sum_{n=0}^{\infty} (-1)^n \left(\frac{2}{z}\right)^n$$

$$\text{So, } \frac{1}{5(z+2)} = \frac{1}{5} \sum_{n=0}^{\infty} (-1)^n \frac{2^n}{z^{n+1}}$$

expand

$$\frac{-z+2}{5(z^2+1)}$$

$$\frac{1}{z^2+1} = \frac{1}{z^2} \cdot \frac{1}{1+\frac{1}{z^2}} = \frac{1}{z^2} \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}$$

$$\text{Thus, } \frac{-z+2}{5(z^2+1)} = \frac{1}{5} \left(\frac{-z+2}{z^2} \right) \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}$$

$$= \frac{1}{5} \left(\frac{1}{z} + \frac{2}{z^2} \right) \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}$$

(ii) Laurent expansion for $1 < |z| < 2$

Expand $\frac{1}{z+2}$

$$\frac{1}{z+2} = \frac{1}{2} \cdot \frac{1}{1+z/2} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2} \right)^n$$

Hence,

$$\frac{1}{5(z+2)} = \frac{1}{10} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z}{2} \right)^n$$

Expand

$$\frac{-z+2}{5(z^2+1)}$$

since

$$|z| > 1$$

$$\frac{1}{z^2+1} = \frac{1}{z^2} \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}$$

Thus,

$$\frac{-z+2}{5(z^2+1)} = \frac{1}{5} \left(\frac{1}{z} - \frac{2}{z^2} \right) \sum_{n=0}^{\infty} (-1)^n \frac{1}{z^{2n}}$$

expand

$$f(z) = \frac{z+3}{z(z^2-z+2)}$$

1. $|z| < 1$

2. $1 < |z| < 2$

$$f(z) = \frac{z+3}{z(z^2-z+2)}$$

$$\frac{z+3}{z(z-3)(z+1)} = \frac{A}{z} + \frac{B}{z-1} + \frac{C}{z+1}$$

after solving:

$$\frac{z+3}{z(z-2)(z+1)} = -\frac{3}{2z} + \frac{2}{z+1} - \frac{1}{2(z-2)}$$

(i) Expansion for $|z| < 1$

Expand $\frac{2}{z+1}$

$$\frac{2}{z+1} = 2 \sum_{n=0}^{\infty} (-1)^n z^n$$

expand $\frac{-1}{2(z-2)}$

$$= \frac{-1}{2(z-2)} = \frac{1}{4} \sum_{n=0}^{\infty} (z/2)^n$$

$$f(z) = -\frac{3}{2z} + 2 \sum_{n=0}^{\infty} (-1)^n z^n + \frac{1}{4} \sum_{n=0}^{\infty} (z/2)^n$$

3 Evaluate the integral

$$\int_0^{2\pi} \frac{d\theta}{5-3\cos\theta}$$

Standard formula:

$$\text{Put } z = e^{i\theta} \Rightarrow d\theta = \frac{dz}{iz}$$

$$\int_0^{2\pi} \frac{d\theta}{a+b\cos\theta} = \frac{2\pi}{\sqrt{a^2-b^2}}, \quad a > |b| \quad \cos\theta = \frac{1}{2}\left(z + \frac{1}{z}\right)$$

Here,

$$a = 5, \quad b = -3$$

So,

$$I = \frac{2\pi}{\sqrt{5^2 - 3^2}} = \frac{2\pi}{4} = \frac{\pi}{2} \quad \text{Ans}$$

18/8/26